Group 5:
Power System, Negative Spot Prices and Flexibility

CIGRE NGN HACKATHON 2026 | Data of 6 April 2025

TEAM MEMBERS

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Challenge A

Minimize the cost to produce heat

Objective

We minimise the total daily heat cost of an industrial site by running an electric boiler whenever it is cheaper than gas.

Our idea

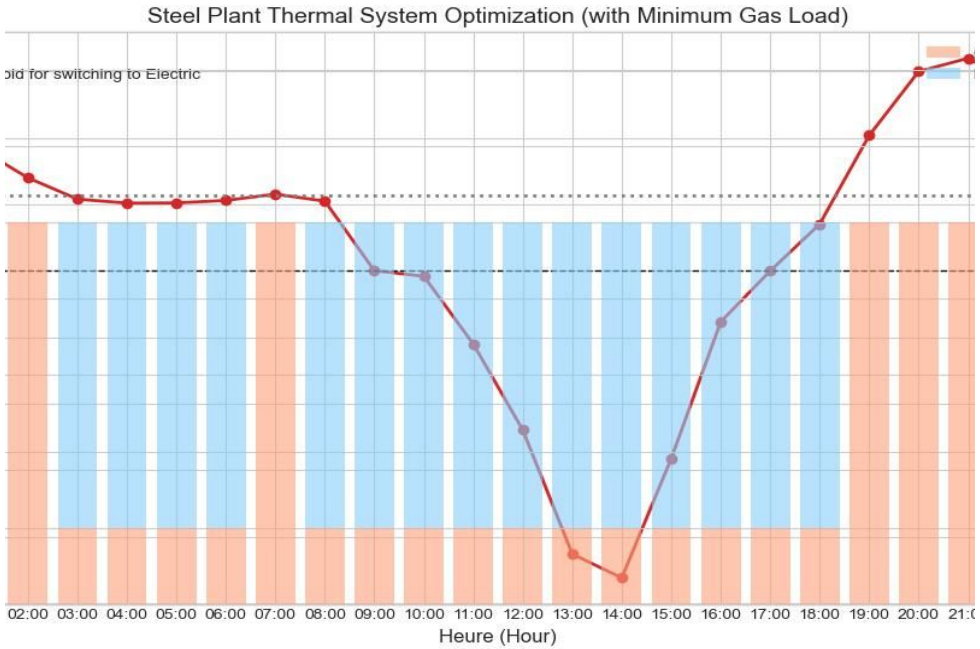
A 50 MW electric boiler beside the gas boiler, switched by the hourly spot price:

- 40 MWth electric in 15 hours
- Gas keeps a 10 MWth minimum



Challenge A

Dual-boiler dispatch on 6 April



Dispatch rule

Electric boiler ON when:

$$(\text{Spot} + 15) \div 0.98 < 40 \div 0.90$$

Assuming spot < 28.55 €/MWh: 40 MWth electric + 10 MWth gas min. Otherwise gas only (50 MWth).

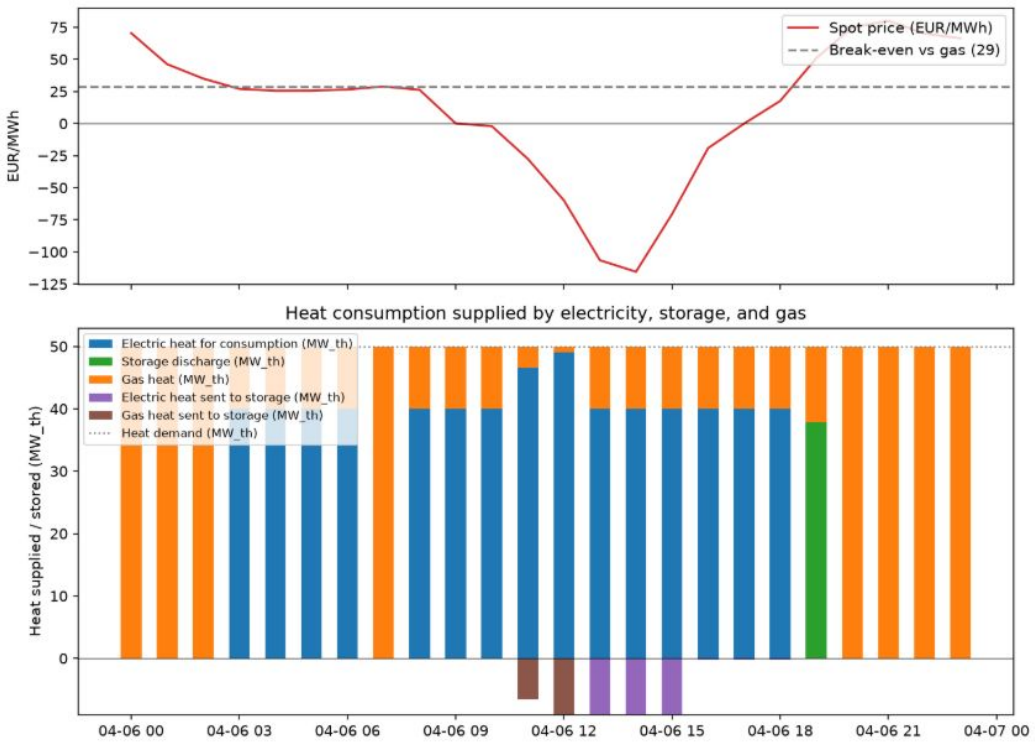
Net daily gain: 26k€

Assumptions

- Constant heat demand 50 MWth
- Gas minimum load 10 MWth
- Grid fee and taxes 15 €/MWh
- Electric boiler 98%, 5 M€, 15 years

Challenge A

Extending the model



Dispatch rule

Store heat where it is cheapest, release it where gas would run.

Stored heat replaces gas at 44.4 €/MWh_th.

Charge in the cheapest hours and discharge in the first gas hour.



+31k€

Net daily gain over a pure gas system

Challenge A

Is a model with thermal storage profitable?

Boiler 50 MWe

Baseline configuration

Saving vs gas only **27,785 €**

Investment **5.0 M€**

Capital cost per day **913 €**

Net daily gain **26,872 €**

Days to cover annual cost **12.0 days**

OPTIMAL ROI

50 MWe + 40 MWh

Medium storage solution

Saving vs gas only **32,354 €**

Investment **5.6 M€**

Capital cost per day **1,023 €**

Net daily gain **31,331 €**

Days to cover annual cost **11.5 days**

Challenge A

Who pays, and the limits

Operator
Invests 5.0M€, gains 27–31 k€ on the day, before fixed costs.

Generators selling at negative prices
Fund 16.4 k€ of the gain (20k€ with storage).

Gas supplier
Sells 667–755 MWh less.

Grid users
Effect on network costs not modelled.

KEY EXPOSURE TAKEAWAY

92%

of the total gain

25.7 of 27.8 k€ comes directly from the **8 negative-price hours**.

The overall business case depends on who is exposed to those specific market prices.

Challenge B

Building Flexibility Stack

Pumped hydro (STEP)

Power: ~4.5 GW pumping **| Response:** 2–5 min
Constraint: Reservoir state of charge, site limits
Actors: EDF Hydro, CNR, RTE

Batteries

Capacity: 1.7 GWh **| Response:** < 1 s · 2–4 h
Constraint: State of charge, cell degradation
Actors: 80% SoC · Developers, RTE

Interconnections

Power: Set by neighbours **| Response:** Instant · continuous
Constraint: Neighbours also in solar surplus (Germany, Spain)
Actors: Price spreads, market coupling · RTE, TSOs

Demand response

Power: 3.0–3.5 GW **| Response:** 30–60 min · 2–4 h window
Constraint: Process continuity, comfort, software
Actors: Bill savings via tariffs/rebates · Industry, aggregators

Nuclear modulation

Cut: 8 GW observed **| Rate:** ~1 GW/h avg · 8–12 h
Constraint: Xenon, boron shim, power limit
Actors: Lost output vs paying negative prices · EDF, RTE

Solar and wind curtailment

Power: Up to 23.7 GW peak **| Response:** < 1 min solar, 5-10min wind
Constraint: Downward only; weather-dependent
Actors: Contract-dependent; paid if forced · RTE, Enedis

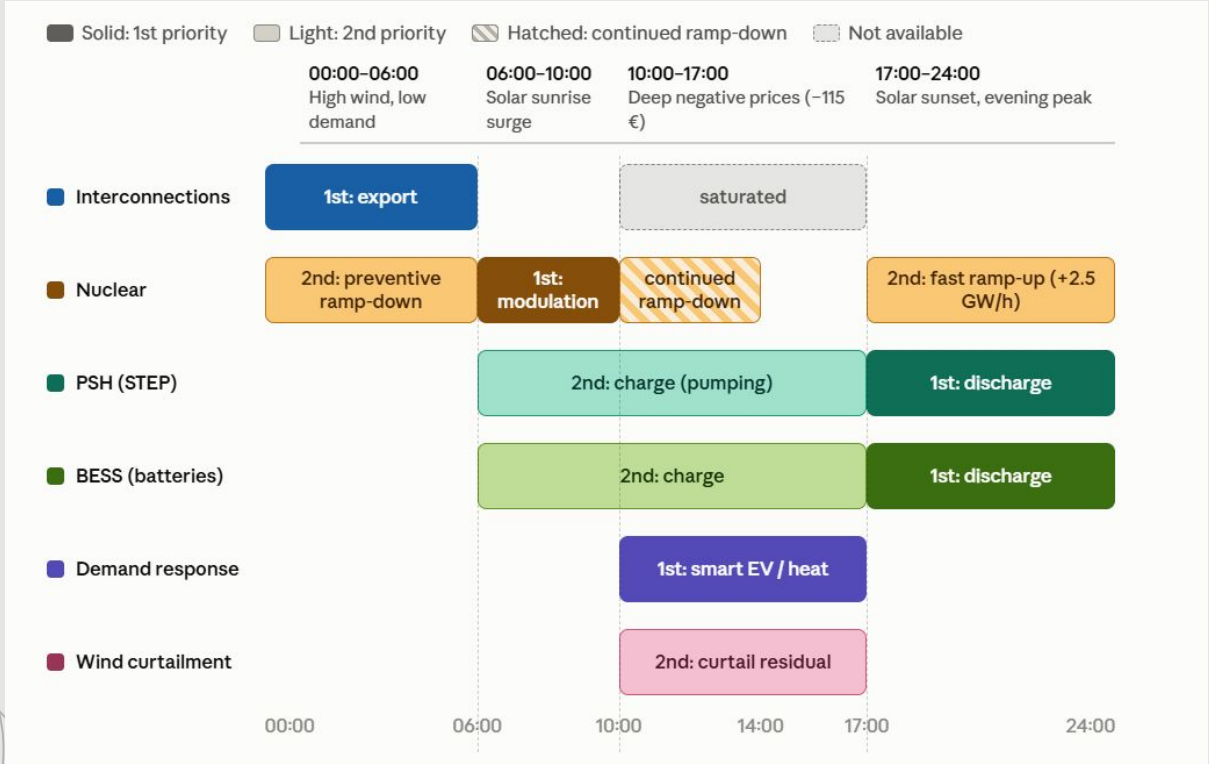
Challenge B

Order and decision rule

1. STEP (Pumped Hydro) With 75–80% efficiency. Absorbs power with almost no cost and stores energy for evening peak hours.
2. BESS (Batteries) Provides millisecond-level frequency stabilization and secondary short-duration storage
3. Nuclear Modulation Large power, no compensation to pay, must start 2 h ahead
4. Demand Response Aggregates EV fleet charging, industrial electric boilers, and heat pumps.
5. RES Curtailment It is used strictly when all storage reservoirs, export capacities, demand-side flexibility, and nuclear ramping capabilities are fully saturated.
6. Interconnections Exporting excess energy to neighboring countries.

Challenge B

Activation plan, 6 April



At night, with strong wind and low demand, we export the surplus and start a preventive nuclear ramp-down. In the morning, as solar rises, we reduce nuclear early because it is slow, and begin charging storage. Between 10:00 and 17:00, exports saturate, to prevent negative prices, we cut wind output and keep ramping nuclear down until about 14:00, so that producers stop selling at a loss, and we absorb the rest locally with smart EV and heating demand and STEP/BESS charging. In the evening, we release the stored energy through STEP and BESS to cover the peak while nuclear ramps back up

Challenge B

Bonus:Power-to-Hydrogen (Electrolysis)

Strategic Positioning:

- **Not in the Core 6 Levers:** Beyond the standard Challenge B toolkit, representing an innovative long-term flexibility vector.
- **2030+ Horizon:** Currently limited in grid-scale capacity, but serves as the ultimate long-duration storage solution for the 2030 energy transition.

Why Power-to-Hydrogen?

- **Sinks Extreme Surplus:** Consumes zero/negative-cost solar electricity during midday hours (10:00–17:00) to produce green H2
- **De-link Storage Limits:** Unlike batteries or STEP (which hit MWh capacity limits), hydrogen can be injected into the gas grid or stored seasonally.

Engineering Assessment:

- **Recommended Role:** Easy to size dynamically based on negative price signals; ideal for absorbing residual curtailed renewable energy in future 2030 grid scenarios.

Thank you for your attention!